WORKED SOLUTION

STEP II 2000, Question 8

Pure • Separable Differential Equations

STEP II 2000, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q8

The key idea

Both equations are separable: the x-dependence and the y-dependence come as a product, so we can put everything involving y on one side and everything involving x on the other. Each x-side has the form $x\,\mathrm{e}^{ax^2}$, which integrates at once because x is (a multiple of) the derivative of x^2. In part (ii) the power of $(y+3)$ is ours to choose, and we choose it so that the growth of y exactly matches e^{3x^2}.

Solution

Part (i)

Since $y = 6$ at $x = 0$, we have $y + 3 > 0$ near the start, and we may divide by $(y+3)^{1/2}$:

$$(y+3)^{-1/2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -4x\,\mathrm{e}^{-x^2}.$$

Integrating both sides with respect to x, and using $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{-x^2} = -2x\,\mathrm{e}^{-x^2}$,

$$2(y+3)^{1/2} = 2\mathrm{e}^{-x^2} + C.$$

Putting $x = 0$, $y = 6$ gives $2 \cdot 3 = 2 + C$, so $C = 4$ and

$$(y+3)^{1/2} = \mathrm{e}^{-x^2} + 2.$$

The right-hand side is always at least 2, so it is a legitimate (positive) square root and $y + 3$ never reaches 0; the division above is justified for all $x \geq 0$. Squaring,

$$\boxed{y = (2 + \mathrm{e}^{-x^2})^2 - 3 = 1 + 4\mathrm{e}^{-x^2} + \mathrm{e}^{-2x^2}}.$$

As $x \to \infty$, $\mathrm{e}^{-x^2} \to 0$ and $\mathrm{e}^{-2x^2} \to 0$, so $y \to 1$, as required.

Part (ii)

Here the equation is

$$(y+3)^{k-1}\,\frac{\mathrm{d}y}{\mathrm{d}x} = x\,\mathrm{e}^{6x^2}.$$

For a general (possibly non-integer) power to make sense we need $y+3>0$, and since $\frac{\mathrm{d}y}{\mathrm{d}x}>0$ for $x>0$, y increases, so $y+3$ stays positive once it starts positive. For $k\neq 0$, integrating gives

$$\frac{(y+3)^k}{k} = \frac{\mathrm{e}^{6x^2}}{12} + C, \qquad \text{so} \qquad (y+3)^k = \frac{k}{12}\,\mathrm{e}^{6x^2} + kC.$$

Ask yourself Which k should we try?

If $k>0$, then for large x the constant is negligible and $y+3 \approx (k/12)^{1/k}\,\mathrm{e}^{6x^2/k}$. We want y to grow like e^{3x^2}, so we need $6/k=3$, that is, $k=2$. (Any other $k>0$ makes $\mathrm{e}^{-3x^2}y$ tend to 0 or to ∞.)

So take $k=2$. Then the equation is $\frac{\mathrm{d}y}{\mathrm{d}x} = x\,\mathrm{e}^{6x^2}/(y+3)$, and the solution is

$$(y+3)^2 = \frac{\mathrm{e}^{6x^2}}{6} + D,$$

where $D=(y_0+3)^2-\frac{1}{6}$ and y_0 is the value of y at $x=0$. The right-hand side increases from $(y_0+3)^2>0$, so it stays positive and the solution exists for every $x\geq 0$:

$$y = \sqrt{\tfrac{1}{6}\mathrm{e}^{6x^2}+D}\;-3.$$

Multiplying by e^{-3x^2} and taking $\mathrm{e}^{-3x^2}=\sqrt{\mathrm{e}^{-6x^2}}$ inside the root,

$$\mathrm{e}^{-3x^2}y = \sqrt{\tfrac{1}{6}+D\,\mathrm{e}^{-6x^2}}\;-\;3\mathrm{e}^{-3x^2} \longrightarrow \sqrt{\tfrac{1}{6}} = \frac{1}{\sqrt{6}}$$

as $x\to\infty$, because both exponentials tend to 0. So with $k=2$ the limit is

$$\boxed{\lim_{x\to\infty}\mathrm{e}^{-3x^2}y = \frac{1}{\sqrt{6}}},$$

whatever the initial value: the constant D has disappeared from the answer, which is why the question can say "any solution".

When $k=2$ the power $(y+3)^{-1}$ also makes sense for $y<-3$. On that branch $y+3=-\sqrt{\frac{1}{6}\mathrm{e}^{6x^2}+D}$, and the same argument gives the limit $-1/\sqrt{6}$. The intended answer is $1/\sqrt{6}$, for solutions with $y>-3$.

Checks

- In (i), $x = 0$ gives $y = (2+1)^2 - 3 = 6$, and differentiating $y = (2 + \mathrm{e}^{-x^2})^2 - 3$ gives $-4x\mathrm{e}^{-x^2}(2 + \mathrm{e}^{-x^2}) = -4x\mathrm{e}^{-x^2}(y+3)^{1/2}$.
- Stepping both equations numerically (Runge–Kutta) agrees with the formulae; for example in (i) $y(2) \approx 1.07360$, and in (ii), starting from $y_0 = 6$, $\mathrm{e}^{-12}y(2) \approx 0.408230$, already close to $1/\sqrt{6} \approx 0.408248$.

Common slips

- Forgetting the constant of integration until after squaring, or fitting $y = 6$ to the wrong expression.
- Integrating $x\,\mathrm{e}^{6x^2}$ as $\mathrm{e}^{6x^2}/6$ rather than $\mathrm{e}^{6x^2}/12$; this does not change k, but it does change the limit.

Beyond STEP

Part (ii) is a small piece of asymptotic analysis: we found how a solution behaves for large x without caring about its initial value, because the constant of integration is swamped by the growing term. Asking which features of a solution are independent of initial conditions is central to the qualitative theory of differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers separable equations and this style of long-term behaviour.

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